

Linear Algebra

Miscellaneous

- $\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}$ and $\sin \theta = \frac{\|a \times b\|}{\|a\| \|b\|}$
- NOTE: $\|v\| = \sqrt{v^T v}$. Don't forget the square root!
- The inverse of a matrix is unique.

Kinds of Matrices

- **Symmetric Matrix:** $A^T = A$
 $\rightarrow A^T A$ and $A A^T$ are symmetric and PSD. It follows that $A^T A + nI$ is PSD and invertible.
- **Orthogonal Matrix:** $Q^T = Q^{-1}$. The columns of Q are orthonormal vectors.
 $\rightarrow \|Qx\| = \|x\|$ - multiplication doesn't change magnitudes or dot products!
 $\rightarrow$ Because $I = Q^T Q = Q Q^T$, implying the rows are also orthonormal!
- **Observation:** Let A be a data matrix with each column matrix representing data from some observation. Then $A^T A$ is a correlation matrix.
- **Covariance Matrix:** Any symmetric and PSD matrix. You can generate the corresponding [random vector](#).

GS 2.6 LU Decomposition

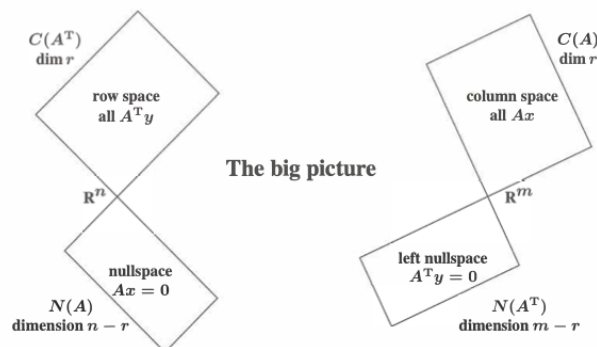
Gauss Jordan Elimination turns a matrix A into an upper triangular matrix U , and the translations needed to accomplish this can make up a lower triangular matrix L , which ultimately leads to the $A = LU$ decomposition.

Dividing out the diagonal values to turn them into 1s can be done with a diagonal matrix D for the LDU decomposition

GS 2.7 Subspaces

The rank of a $m \times n$ matrix is the rank/dimension of the column space (and of the row space since they are equal) of the matrix.

Also the nullspace of matrix A is x s.t. $Ax = 0$ and the left nullspace is $x^T A = A^T x = 0$



- The rowspace and nullspace are orthogonal complements ($\text{span}(A^T), N(A)$)
- So are the column space and the nullspace of the transpose ($\text{span}(A), N(A^T)$)

GS 4.2 Projection

Projecting a vector b onto the line through vector a is easy. Call the projection $p = \tilde{x}a$, then the component $b - p = b - \tilde{x}a$ is perpendicular to a , so we have

$$a \cdot (b - \tilde{x}a) = 0 \quad \tilde{x} = \frac{a \cdot b}{a \cdot a} = \frac{a^T b}{a^T a} \quad p = a\tilde{x} = \frac{aa^T}{a^T a} b = \frac{aa^T}{\|a\|^2} b$$

Projection of b onto the subspace spanned by the basis of column vectors $a_1, \dots, a_n$ (which we condense to matrix A) is equivalent to finding a linear combination of the basis vectors with coefficients $\tilde{x}$ as $A\tilde{x}$.

Then $b - A\tilde{x}$ is perpendicular to all the a_i , and so is in the left nullspace of A .

$$A^T(b - A\tilde{x}) = 0 \quad \tilde{x} = (A^T A)^{-1} A^T b$$

Since we ultimately want the projection of b onto A , we need $p = A\tilde{x} = A(A^T A)^{-1} A^T b$.

Lastly, $A^T A$ is invertible iff A has linearly independent columns (pg. 212), which is proved by showing they have equivalent nullspaces. For example, if $(A^T A)x = 0$, we can write $x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 = 0$

GS 4.3 Least Squares Regression

Suppose we have a matrix $Ax = b$ with no solution. This can happen because there are too many equations in the original system - i.e. too many rows.

Luckily we know that $A^T A$ is invertible iff A has linearly independent columns from 4.2. Alternatively, we can also just show they have the same rank, and so if A is invertible, so is $A^T A$.

So we can multiply both sides by A^T : $A^T A x = A^T b$. This ends up producing the solution which minimizes the error $e = b - Ax$!

Suppose we are trying to fit data points (X, Y) . Then this becomes $X\beta = Y$, and we augment the X coordinates with a column of 1s to leave room for a y -intercept:

$$X = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \quad X^T X = \begin{bmatrix} m & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \quad X^T Y = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Differentiating $\|Ax - b\|$ (or equivalently $\|X\beta - Y\|$) with respect to each component of the n rows of the columns of A produces n equations (from the original $m \times n$ matrix) that we want to minimize (set equal to 0). These end up being equivalent to $A^T A x = A^T b$! (6.036 notes ch7.2).

This is all ok when $A^T A$ is invertible (when it has linearly independent columns). When this isn't the case we have ambiguity in our solution. We'll resolve this later with the pseudoinverse (7.4).

Alternatively, we can add in ridge regularization which yields invertible $A^T A + n\lambda I$ (6.036 notes).

GS 4.4 QR Decomposition and Orthonormal Bases

Recall **Orthogonal Matrix**: $Q^T = Q^{-1}$. The columns of Q are orthonormal vectors.

Permutation, Rotation, Reflection matrices are all orthonormal but whatever.

Projection with an orthonormal basis is easy: $p = Q\tilde{x} = QQ^Tb$. The projection matrix simplifies down to QQ^T .

The Gram-Schmidt Process is the obvious method for orthogonalizing a set of vectors $A, B, C, \dots$:

$$\begin{aligned} q_1 &= \frac{A}{\|A\|} & A &= a \\ q_2 &= \frac{B}{\|B\|} & B &= b - (q_1 \cdot b)q_1 = b - \frac{A^T b}{A^T A} A \\ q_3 &= \frac{C}{\|C\|} & C &= c - (q_1 \cdot c)q_1 - (q_2 \cdot c)q_2 \end{aligned}$$

We can apply this orthogonalization to the columns of matrix to obtain the QR decomposition

$$A = QR$$
$$\begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ 0 & q_2^T b & q_2^T c \\ 0 & 0 & q_3^T c \end{bmatrix}$$

Note that $v(q_i^T v)$ is equal to the component of v in the direction of q_i . Multiplying this through column by column of R reproduces the columns of A .

Notably R is upper triangular. We can simplify $A^T A \rightarrow R^T R$, making the LSE for $A\tilde{x} = b$ into $R\tilde{x} = Q^T b$. This is then solvable very quickly with back substitution because of the upper triangular nature of R .

The runtime of Gram-Schmidt is $O(mn^2)$ for an $m \times n$ matrix. The textbook has pseudo-code on pg. 240 detailing the operations of QR decomposition naively (in truth there's more advanced techniques used to speed up the computation).

GS 5.1 Properties of Determinants

- When A is singular (non-invertible), $\det(A) = 0$.
- $\det$ is the product of the pivots (when eliminated to upper triangular etc.)
 - When upper/lower triangular, just multiply the diagonal entries
- $\det(AB) = \det(A) \det(B)$ (so inverse has $1/\det(A)$ and $\det(A^T) = \det(A)$)
- Useful facts
 - Exchanging rows reverses signs. It follows that (having two equal rows/columns, $\det = 0$) and (subtracting a multiple of one row from another doesn't change $\det(A)$).
 - If two rows are equal, then $\det = 0$
- The determinant is a linear function on each row (multiply a row by a constant and the determinant is multiplied by that constant, create a new matrix that has another row added to the i th row and it's the same as summing the determinants of two matrices with each given row in the i th position)
 - Multiplying a row by a constant k multiplies $\det$ by k
 - Multiply a matrix by a constant k multiplies $\det$ by k^n
 - Adding a multiple of one row (i) to another row (j) doesn't change $\det$
 - * Create a new matrix with a copy of row i in the j th row
 - * Because there are two identical rows, this has $\det = 0$
 - * Multiply the j th row of the new matrix by a constant factor if you desire
 - * Add the two matrices, adding the two determinants
 - Thus to compute determinant we can row reduce

Basis and Invertibility Facts

For $n \times n$ matrix A :

- If a matrix is square with linearly independent columns $\rightarrow$ invertible
- Using the complex space we are guaranteed n eigenvalues, but not n L.I. eigenvectors

Miscellaneous

- Trace is cyclic (verify by bashing):

$$\text{tr}(AB) = \text{tr}(BA)$$

- The inverse of a symmetric matrix, if it exists, is symmetric as well ([ref](#) and [ref](#))

GS 6.1 Introduction to Eigenvalues

Eigenvectors are x s.t. $Ax = \lambda x$ for some scalar x , and λ is the corresponding eigenvalue.

- $A^n = \lambda^n x$ and $A^{-1}x = \lambda^{-1}x$
- Can rearrange and solve $(A - \lambda I)x = 0$
 - $A - \lambda I$ is singular and has no inverse (so its determinant is 0).
 - Equivalent to finding λ such that $\det(A - \lambda I) = 0$.
 - * $\det(A - \lambda I)$ is the **Characteristic Polynomial**
 - * $\det(A - \lambda I) = (z - \lambda_1)^{d_1} \cdots (z - \lambda_m)^{d_m}$, where d_i is the algebraic multiplicity.
 - * Eigenvalues/the roots can be complex!
 - Then use elimination or other tools to solve $(A - \lambda I)x = 0$
 - * The rows of $A - \lambda I$ might have some nice pattern (in 2×2 they're [possibly] multiples of each other with the multiplicative factor being the i th coefficient of the corresponding eigenvector)

Symmetric matrices have **orthogonal eigenvectors**. Allegedly this is explained more later.

Because Singular matrices have nonempty nullspaces, 0 is an eigenvalue with the nullspace as its corresponding eigenvectors.

- The **product** of the eigenvalues with multiplicity (i.e. $\lambda_1^{d_1} \cdots \lambda_m^{d_m}$) equals the determinant of the matrix
 - Simply set $\lambda = 0$ in the expansion of $\det(A - \lambda I)$
- The **sum** of the eigenvalues with multiplicity (i.e. $d_1\lambda_1 + \cdots + d_m\lambda_m$) equals the sum of the n diagonal entries (called the **trace**)

(some of) 6.4 - Complex Eigenvalues

For the product of two complex numbers $c_1 c_2$ we have $\overline{c_1 c_2} = \overline{c_1} \cdot \overline{c_2}$, and the same relation holds for the product of matrices. Therefore for real matrix A , $\overline{Ax} = A\overline{x}$ (because $\overline{A} = A$).

Also, we define the norm of a complex vector as $\|c_1\| = c_1 \cdot \overline{c_1}$.

For a matrix with real entries

- if λ is a complex eigenvalue with eigenvector v
then $\overline{\lambda}$ is **also** an eigenvalue with eigenvector $\overline{v}$

We can simply conjugate both sides of $Ax = \lambda x$:

$$Ax = \lambda x \quad A\overline{x} = \overline{\lambda}\overline{x}$$

Also, the eigenvalues of a real symmetric matrix are **real** (and we can create a full orthogonal eigenbasis - see Spectral Thm 6.4). To show this, convert $Sx = \lambda x$ and $S\overline{x} = \overline{\lambda}\overline{x}$ into

$$\overline{x}^T Sx = \overline{x}^T \lambda x = \overline{x}^T \overline{\lambda} x \quad \overline{x}^T x \geq 0$$

GS 6.2 - Diagonalization

If we have n distinct (linearly independent) eigenvectors for matrix A , we can apply a change of basis to create a diagonalized form:

- Using eigenvectors $x_1, \dots, x_n$ and corresponding eigenvalues $\lambda_1, \dots, \lambda_n$
- Make change of basis matrix $X = [X_1, \dots, X_n]$ using column vectors x_i (converts from eigenvector basis back to the standard basis). X^{-1} intuitively does the reverse.
- Make diagonal matrix $\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$ and we have $A = X\Lambda X^{-1}$

We can use this to compute powers of a matrix very easily since Λ is diagonal:

$$A^k = X\Lambda X^{-1} \cancel{X} \dots \cancel{X^{-1}} X\Lambda X^{-1} = X\Lambda^k X^{-1} = X \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} X^{-1}$$

We can use this technique to compute A^k where k can be as big as we want, for example ∞ when we try to find the stationary state of a markov chain.

We can also generalize this to compute $A^k u$ for some u which we can break down with an eigenbasis:

$$\begin{aligned} A^k u &= A^k (c_1 x_1 + \dots + c_n x_n) \\ &= c_1 A^k x_1 + \dots + c_n A^k x_n \\ &= c_1 \lambda_1^k x_1 + \dots + c_n \lambda_n^k x_n \end{aligned}$$

Matrix invertibility and diagonalizability aren't the same. If any eigenvalue is 0, then the nullspace is nonempty and the matrix is not invertible. But a matrix can still be diagonalizable with an eigenvalue of 0.

Note: Eigenvectors corresponding to different eigenvalues are linearly independent (pg. 306)

We define **similar** matrices as those sharing a central matrix C in a more general change of basis BCB^{-1}

- These matrices all share the same eigenvalues - those of C (pg. 308)
- Comes from generalizing the family of matrices sharing Λ in their diagonalization with different eigenvector matrices X .

-
- **Algebraic Multiplicity:** The power of λ_i in $\det(A - \lambda I)$
 - **Geometric Multiplicity:** The dimension of the eigenspace for λ_i

GS 6.4 - Symmetric Matrices

Symmetric matrices have two critical properties:

- They have only real eigenvalues (if the matrix is real)
- The total eigenvector multiplicity is n .
- The n eigenvectors can be chosen to be orthonormal.

Spectral Theorem: Every symmetric matrix has factorization $S = Q\Lambda Q^T$ with real eigenvalues Λ and orthonormal eigenbasis Q .

Note: Eigenvectors of a real symmetric matrix with different λ are perpendicular (pg. 340)

The pivots and eigenvalues of **symmetric** matrices have the same signs

- The number of positive eigenvalues equals the number of positive pivots etc.
- Also true for 0 eigenvalues since 0 pivots correspond to nullspace!

Note: We are always guaranteed a full basis of eigenvectors, even if there are repeat eigenvalues (pg. 343). There are a few ways to prove this, including using Schur's Theorem.

For non-symmetric matrices, λ may be complex. In this case, complex eigenvalues come in conjugate pairs (and so do their corresponding eigenvectors)

GS 6.5 - Positive Definite Matrices

Matrices with only positive eigenvalues are **positive definite**.

Matrices with all eigenvalues ≥ 0 are **positive semi-definite**.

We can quickly check for this property by row reducing and checking that the pivots are positive (the signs of the pivots are the same as the signs of the eigenvalues)

Intuition for this definition comes from the energy definition:

- $Sx = \lambda x$ implies $x^T Sx = \lambda x^T x = \lambda \|x\|^2$
- So when all $\lambda \geq 0$, we have that $v^T S v \geq 0$ for all v

Note: If S and T are symmetric positive definite, so is $S + T$. (pg. 352)

- just expand $v^T (S + T)v$ to see that both components are always positive
- similarly $A^T A$ and AA^T are always PSD (try $v^T A^T A v = \|Av\|^2 \geq 0$)

Ultimately symmetric matrices with any of the following properties have them all:

- All n pivots positive
- All n upper left determinants positive
- All n eigenvalues of S positive (positive definite)
- $x^T S x > 0$ for all $x \neq 0$
- $S = A^T A$ for a matrix A with independent columns (just like in 4.3! A can be rectangular.)

A useful property of PSD matrices is that the overall determinant is positive (which follows from the above regardless)

One important application of PSD is in minimization of a function - the first derivative must be 0, and the Hessian (second derivative matrix must be positive definite).

GS 7.1 SVD Intro for Matrix Compression

Given vectors u ($m \times 1$) and v ($n \times 1$), then we can generate a rank 1 matrix with uv^T ($m \times n$).

- This allows you to compress mn values down into $m + n$ values!

We can choose these vectors with more nuance and scaling terms to reconstruct a higher rank matrix: $A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T$.

We can choose v_i as the eigenvectors of $A^T A$ and u_i as the eigenvectors of AA^T . Both of these matrices are symmetric.

- $A^T A$ matrices are PSD! Observe $x^T A^T A x = \|Ax\|^2 \geq 0$. So all eigenvalues are ≥ 0 .

We can specifically choose u_i such that $Av_i = \sigma_i u_i$, for the given eigenvalue $\lambda_i = \sigma_i^2$.

- By this setup, u_i is a [unit](#) (see 4.) vector with many eigenvalues
- Using $A^T A v_i = \lambda_i v_i$, we can show Av_i is an eigenvector of AA^T with eigenvalue λ_i .

Ultimately we can write this out as the **Singular Value Decomposition**:

$$A = U \Sigma V^T$$

where U and V are orthogonal matrices of eigenvectors of AA^T and $A^T A$ as described above. The eigenvalues in Σ are sorted in decreasing order down the diagonal. So the first few columns of U, V end up being the most important components of A .

GS 7.2 Bases and Matrices in SVD

We can use the following basis vectors:

- $u_1, \dots, u_r$ is the eigenbasis of AA^T , $u_{r+1}, \dots, u_m$ is the basis of the left nullspace $N(A^T)$
- $v_1, \dots, v_r$ is the eigenbasis of $A^T A$ and $v_{r+1}, \dots, v_n$ is the basis of the nullspace $N(A)$
- The matrices [share](#) the same nonzero eigenvalues!
- We construct u from v via $Av_i = \sigma_i u_i$ (proof linked above).

→ We can expand this to $AV = \Sigma U$, which rearranges to the SVD.

We generalize the aforementioned decomposition: $A = \sigma_1 u_1 v_1^T + \dots + \sigma_n u_n v_n^T$

- This corresponds to the “full” svd which has many zero entries.
- Removing the zero'd σ_n terms, we get a new Σ which is just $r \times r$, instead of $m \times n$

A [complete and very helpful derivation](#) with many details can be accessed [here](#)!

The singular value decomposition is equivalent to just diagonalizing the matrix if it's PSD (and by definition symmetric). However the singular values are always positive and real, which may not be the case for the diagonalization.

Eckart-Young Theorem: The approximation $A' = \sum_{i=1}^k \sigma_i u_i v_i^T$ is the rank k approximation matrix that minimizes the matrix Frobenius norm (sum of squared entries) $\|A - A'\|_F$

GS 7.3 PCA by the SVD

The way we typically organize matrices when talking about PCA is a little different - each subject gets a row of its measurements. This is like how we organize data in Linear Regression. Strang's description of PCA is a bit sparse so these notes also contain bits of [this video](#).

1. Suppose we have a $n \times 15$ matrix of experimental observations $\mathbb{X}$ (i.e. we have 15 measurements for each of the n items).
2. Define the mean row $\tilde{x} = \frac{1}{n} \sum x_i$, and balance each column by subtracting the mean row:

$$B = \mathbb{X} - \tilde{\mathbb{X}} = \begin{bmatrix} - & x_1 & - \\ & \vdots & \\ - & x_n & - \end{bmatrix} - \begin{bmatrix} - & \tilde{x} & - \\ & \vdots & \\ - & \tilde{x} & - \end{bmatrix}$$

3. Then $C = B^T B$ is a 15×15 symmetric matrix
 - Dividing by $n - 1$ we get the covariance matrix $E[(M_i - E[M_i])(M_j - E[M_j])]$
4. Diagonalizing gives $CV = V\Lambda$ where V is the orthogonal matrix of eigenvectors.
5. The principal components T are then $T = BV$
 - These principal components span the column space of B
 - The i th principal component has contribution $\frac{\lambda_i}{\sum \lambda_i} = \frac{\sigma_i^2}{\sum \sigma_i^2}$
 - Alternatively, let $B = U\Sigma V^T$, then $T = BV = U\Sigma$

It can be hard to parse this all. Essentially

1. We find the eigenvectors V of the (15×15) covariance matrix $C = B^T B$.
2. These become the principal directions v_i of the PCA - v_i accounts for variance $\lambda_i = \sigma_i^2$.
 - These are sometimes called "loadings" or "principal directions", Strang calls these directions the **principal components** and calls it a day.
 - On their own, these tell us the axes which we should use to visualize our data, but we haven't actually changed to the new basis yet.
3. We then project each row x_i of B onto v_j to change the basis from the standard m -d basis into the eigenvector basis
 - We use $P_{v_j}(x_i) = v_j \left(\frac{v_j^T x_i}{||v||^2} \right) = v_j(v_j^T x_i)$
 - This boils down to $T = BV$ on the macro scale (see below). These are what other sources (stupidly) refer to as the **"principal components"** (confusing + misleading)

$$T = BV = \underbrace{\begin{bmatrix} - & x_1 & - \\ & \vdots & \\ - & x_n & - \end{bmatrix}}_{n \times m} \underbrace{\begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}}_{m \times m}$$

- In some [implementations](#) V is $m \times \min(m, n)$ since if $n < m$ the $m - n$ extra eigenvalues are 0 so the corresponding eigenvectors are useless anyways.
4. Our data is now reparametrized according to the orthonormal principal directions basis!

GS 7.4 Geometry of the SVD and Pseudoinverses

We define the *norm* of a matrix A as its largest growth rate. This equals largest singular value σ_1 :

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sigma_1$$

Visualizing a matrix using the SVD (rotate, stretch, rotate) helps justify this. Or $Av_1 = \sigma_1 u_1$.

It follows that $\|A + B\| \leq \|A\| + \|B\|$ and $\|AB\| \leq \|A\| \cdot \|B\|$

This section also introduces the *Polar Decomposition* $A = QS$ with orthogonal Q and PSD S .

- when A is invertible, S is directly positive definite.

Essentially we separate a matrix into its rotate (Q) and stretch (S) components (like $e^{i\theta} \cdot r$):

$$A = U\Sigma V^T = (UV^T)(V\Sigma V^T) = (Q)(S)$$

This leads to the useless fact: $Q = UV^T$ is the closest orthogonal matrix to A (Frobenius norm)

Lastly, we have the glorious pseudoinverse (wooooo): $A^\dagger$

Consider first a (tall skinny) $n \times m$ matrix with rank m (linearly independent columns).

- $A^T A$ ($m \times m$) is invertible, so we can use the left inverse $(A^T A)^{-1} A^T$

Alternatively for a (short fat) $n \times m$ matrix with rank n (linearly independent rows)

- AA^T ($n \times n$) is invertible, so we can use the right inverse $A^T (AA^T)^{-1}$

We can use the SVD to generalize this to $r(A) < m, n$. Define $A^\dagger$

$$A^\dagger = V\Sigma^\dagger U^T = V \begin{bmatrix} \sigma_1^{-1} & & \\ & \ddots & \\ & & \sigma_r^{-1} \\ \dots & 0 & \dots \end{bmatrix} U^T$$

Then we can somewhat invert A :

$$A^\dagger A = V\Sigma^\dagger U^T U \Sigma V^T = V\Sigma^\dagger \Sigma V^T = V \begin{bmatrix} \mathbf{I} & 0 \\ 0 & \mathbf{0} \end{bmatrix} V^T$$

Note that the zero'ed columns of the pseudo-identity ($\Sigma^\dagger \Sigma$) correspond to eigenvectors with eigenvalues 0 (and are in the nullspace). So applying the pseudoinverse will drop all components that would be in the nullspace of A^T (and thus the orthogonal complement of its image), and invert the components that are in the image of A .

For example, suppose we are solving $Ax = b$ with $b = Ay + z$ where $z \in N(A^T) = \text{Im}(A)^\perp$, then we would solve $A^\dagger Ax = A^\dagger b$ and obtain $A^\dagger b = y$ as our approximation of x .

Because the pseudoinverse returns the components of b that are in the image of input A (and discards perpendicular components), it ends up solving the OLS *regression* problem.

- If we have infinite solutions (underdetermined) returns the x with minimum $\|x\|_F$ norm
- If we have no solution (overdetermined) returns the x with the minimum LSE $\|Ax - b\|_F$